

LESSON 2-5 • SESSION 1 • PRACTICE • FORM B**Range and IQR**

Learning target – I can find the range and the interquartile range of a data set. · 6.SP.B.5c

REMEMBER

- **Range** = greatest – least. It uses only the two extreme values.
- **IQR** = $Q3 - Q1$. It is the spread of the **middle half** of the data.
- Changing only the greatest value changes the range, **not** the IQR. Missing max = min + range.

WORKED EXAMPLE

Hikes (km): 5, 12, 8, 20, 15, 10, 18.

Step 1 Order: 5, 8, 10, 12, 15, 18, 20.**Step 2** Range = $20 - 5 = 15$ km.**Step 3** $Q1 = 8$, $Q3 = 18$, so $IQR = 18 - 8 = 10$ km.**Answer** Range **15 km**; IQR **10 km****WORDS** range (*rango*) · interquartile range (IQR) (*rango intercuartil*) · variability (*variabilidad*)**PRACTICE***show your work · label units*

- 1. Bike Rides.**
- Seven rides covered 6, 14, 9, 22, 17, 11, 19 miles.

Hint Order first. Range uses the ends; IQR uses $Q1$ and $Q3$.

- a.** Range: ____ **b.** $Q1$: ____ **c.** $Q3$: ____ **d.** IQR: ____

- 2. Car Wash.**
- Seven sales earned 52, 66, 58, 74, 55, 69, 62 dollars.

- a.**
- Range: ____

- b.**
- IQR: ____

- 3. Band Rehearsals.**
- Seven lasted 50, 68, 58, 84, 62, 75, 66 minutes.

Hint Leave the median out of both halves.

- a.**
- Range: ____

- b.**
- IQR: ____

- 4. Swapped Value.**
- A data set is 10, 15, 17, 21, 26, 28, 34. Someone replaces 34 with 90. Darnell says both the range and the IQR grow.

Hint Only the greatest value changed.

- a.**
- Range before; after: ____

- b.**
- IQR before; after: ____

- c.**
- Is he right? Explain.

- 5. Missing Maximum.**
- Seven values have min 9, range 41,
- $Q1 = 18$
- and
- $Q3 = 37$
- .

Hint Add the range to the minimum.

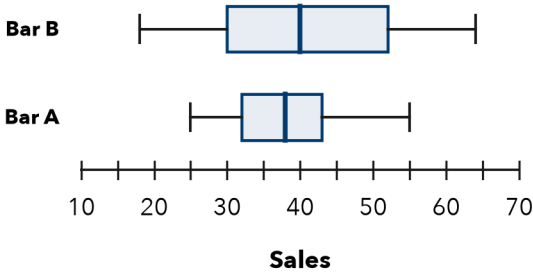
- a.**
- Maximum: ____

- b.**
- IQR: ____

6. **Same Range?** Set C: 10, 15, 20, 25, 30, 35, 40. Set D: 10, 11, 13, 25, 37, 39, 40.
 a. Range of C: ____ b. IQR of C: ____ c. Range of D: ____ d. IQR of D: ____

7. **Which Spread?** Hourly wages: min \$12, Q1 \$15, median \$18, Q3 \$21, max \$70.
 a. Range: ____ b. IQR: ____ c. Better for typical spread: ____

8. **Two Snack Bars.** The box plots show daily sales.
 a. Range and IQR of each bar:
 b. Which bar has the steadier middle half? _____
 c. Why could the ranges mislead if one day had unusual weather?



FIND THE ERROR

explain the mistake, then fix it

9. **Marisol's Work.** For 5, 9, 14, 18, 25, 31, 40, Marisol wrote: "The IQR is 35, the greatest value minus the least." Explain the mistake and find the IQR.

TEST PRACTICE

choose the best answer

10. The IQR describes the spread of the –
 A whole data set B middle half C lowest values D highest values
11. What is the range of 12, 19, 8, 25, 16?
 A 17 B 13 C 8 D 25

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

26 Q	16 M	32 Z	50 U	14 A	19 R	17 S	8 X	10 E	58 E
#1a		#1d		#2b		#3b		#5a	

CHECK YOURSELF

☐ I can do this on my own
 ☐ I need a little help
 ☐ I have a question about #____